

IMMUNOLOGY

A human homolog of SIR2 antiphage proteins mediates immunity via the Toll-like receptor pathway

Delphine Bonhomme^{1†}, Hugo Vaysset^{2,3†},
Eirene Marie Q. Ednacot^{4,5}, Vasco Rodrigues⁶, Yazan Salloum⁷,
Jean Cury², Anny Wang^{4,5}, Axel Benchetrit⁸, Pierre Affaticati⁸,
Veronica Hernandez Trejo¹, Paul Vittot¹, Charlie Bories¹,
Alexis Cornec¹, Jean-Pierre Levrard^{8,9}, Pedro P. Hernández⁷,
Philippe Benaroch⁶, Benjamin R. Morehouse^{4,5,10,11},
Aude Bernheim^{2†*}, Enzo Z. Poirier^{1†*}

Key actors of mammalian immunity originated from bacterial antiphage systems. The full extent of immune system conservation between bacteria and eukaryotes is unknown. Here, we show that the silent information regulator 2 (SIR2) protein domain, present in antiphage systems, plays a role in eukaryotic innate immunity. We identified SIRal, a human protein with a SIRim domain (subtype of SIR2) that plays a pivotal role in the animal Toll-like receptor pathway of innate immunity and protects against bacterial and viral infections. Proteins containing a SIRim domain are found across 19% of eukaryotic genomes, including zebrafish, where SIRal plays a role in inflammation. This work opens up avenues of research on the immune role of eukaryotic SIRim proteins as well as on the involvement of SIRal in human pathology.

The arms race between hosts and pathogens drives continuous diversification of immune mechanisms. Because bacterial, archaeal, and eukaryotic organisms face distinct threats, research thus far has largely focused on clade-specific immunity. However, recent work has uncovered immune mechanisms that are shared between bacteria and eukaryotes, including humans, revealing an overlooked aspect of immunity: conservation (1–3). This conservation includes the cGAS-STING pathway—which is pivotal in the detection of DNA viruses—as well as caspases and gasdermins, which regulate cell death, and viperins, which are central antiviral effectors (4–9). Similarly, Toll-interleukin-1 receptor (TIR) domains are involved in antiphage systems and animal immune pathways (10–12). In prokaryotes and plants, TIR domains perform an enzymatic function, degrading NAD⁺ (oxidized form of nicotinamide adenine dinucleotide) to trigger cell demise or to produce immune secondary messengers (10, 12–14). In mammals, Toll-like receptors (TLRs), which are crucial in detecting pathogen cues upon infection, transduce immune signals via a cytosolic TIR domain (11). The breadth of conservation of immune actors across domains of life remains to be investigated.

Prokaryotes and eukaryotes encode proteins bearing a silent information regulator 2 (SIR2) domain, originally described in the sirTuin family of enzymes (15). By performing NAD⁺-dependent deacetylation, sirTuins play an important role in regulating gene expression through

the modification of histones in eukaryotes, or through the regulation of metabolism in bacteria (15, 16). Recently, a previously unknown function was uncovered for the SIR2 domain in bacteria: a role in antiphage defense (12, 17–20). Multiple antiphage systems encode a SIR2 domain that provides immunity through degradation of cellular NAD⁺. For example, phage-mediated activation of the protein Thoeris A (ThsA) leads to SIR2-dependent depletion of NAD⁺ and growth arrest of the infected cell (12, 19, 20). SIR2 domain-containing proteins thus perform immune and nonimmune functions in bacteria. Whether SIR2 domains from antiphage systems are conserved in eukaryotes and contribute to human immunity is presently unknown.

Results

Identification of SIRal, a human homolog of SIRim antiphage proteins

We set out to identify all prokaryotic and eukaryotic proteins containing a SIR2 domain encoded in publicly available sequenced genomes. We first implemented homology-based detection of SIR2 proteins that are unrelated to bacterial, archaeal, or eukaryotic immunity and of SIR2 proteins involved in bacterial antiphage systems. To do so, we built a general protein profile for the SIR2 family using 4302 SIR2 homologs from across the tree of life (see Materials and methods) and used it to search a database of 30,254 bacterial, archaeal, and eukaryotic genomes. This homology search identified 57,911 SIR2 proteins across kingdoms (29,634 in bacteria, 2094 in archaea, and 26,183 in eukaryotes; table S1). To better characterize the diversity within the SIR2 family, we constructed a phylogenetic tree (Fig. 1A and Materials and methods). In this tree, SIR2 proteins are clustered into two monophyletic groups (Fig. 1A, beige and brown), with each group composed of bacterial, archaeal, and eukaryotic proteins (inner circle: blue, gray, and green, respectively). This indicates that the family of SIR2 proteins can be partitioned into two subfamilies of ancient origin.

To better characterize the two subfamilies, we mapped the proteins involved in bacterial immunity on the phylogeny. We first identified bacterial SIR2 proteins involved in documented antiphage systems with the program DefenseFinder (Fig. 1A, outer circle) (21). All bacterial antiphage SIR2 proteins are located in one of the two clades, suggesting that the entire clade may be involved in immunity (Fig. 1A, brown clade). By contrast, all the bacterial and eukaryotic nonimmune sirTuins are found in the other clade (beige clade) (22). Because DefenseFinder only identifies homologs of defense systems whose antiphage activity has been experimentally validated, these predictions likely represent only a fraction of all existing antiphage systems. We thus sought to identify additional putative SIR2-containing antiphage proteins. To do this, we leveraged the tendency of antiphage genes to cluster within bacterial genomes and analyzed the frequency with which known antiphage systems are found in the vicinity (± 20 genes) of SIR2 proteins (18, 23, 24). The genomic neighborhood of bacterial SIR2 proteins from the first, brown clade shows a 12-fold enrichment in antiphage systems compared with the neighborhood of SIR2 proteins from the second, beige clade (Fisher's exact test, $F = 12.40$, $p < 10^{-3}$; Fig. 1A and fig. S1). Such bipartite distribution strongly suggests that most of the bacterial proteins from the first clade, but not from the second clade, have an immune function (18, 23, 24). This phylogenetic analysis led us to divide the SIR2 family of proteins into two subfamilies with seemingly distinct functions: (i) sirTuins, which have a classical SIR2 domain and are involved in nonimmune functions (Fig. 1A, beige clade), and (ii) sirims, which are equipped with a SIRim domain (for SIR and immunity, described hereafter) and are present in validated

¹Innate Immunity in Physiology and Cancer Team, Institut Curie, PSL Research University, INSERM U932, Paris, France. ²Molecular Diversity of Microbes Lab, Institut Pasteur, Université Paris-Cité, CNRS UMR3525, Paris, France. ³AgroParisTech, Université Paris-Saclay, Paris, France. ⁴Department of Molecular Biology and Biochemistry, School of Biological Sciences, University of California Irvine, Irvine, CA, USA. ⁵Department of Pharmaceutical Sciences, School of Pharmacy and Pharmaceutical Sciences, University of California Irvine, Irvine, CA, USA. ⁶Myeloid Cells and Immunity Team, Institut Curie, PSL Research University, INSERM U932, Paris, France. ⁷Institut Curie, PSL Research University, CNRS UMR 3215, INSERM U934, Paris, France. ⁸TEFOR Paris-Saclay, CNRS UAR2010, Université Paris-Saclay, Saclay, France. ⁹Université Paris-Saclay, CNRS, Institut Pasteur, Université Paris-Cité, Institut des Neurosciences Paris-Saclay, Saclay, France. ¹⁰Institute for Immunology, University of California Irvine, Irvine, CA, USA. ¹¹Center for Virus Research, University of California Irvine, Irvine, CA, USA. *Corresponding author. Email: enzo.poirier@curie.fr (E.Z.P.); aude.bernheim@pasteur.fr (A.Ber.) †These authors contributed equally to this work.

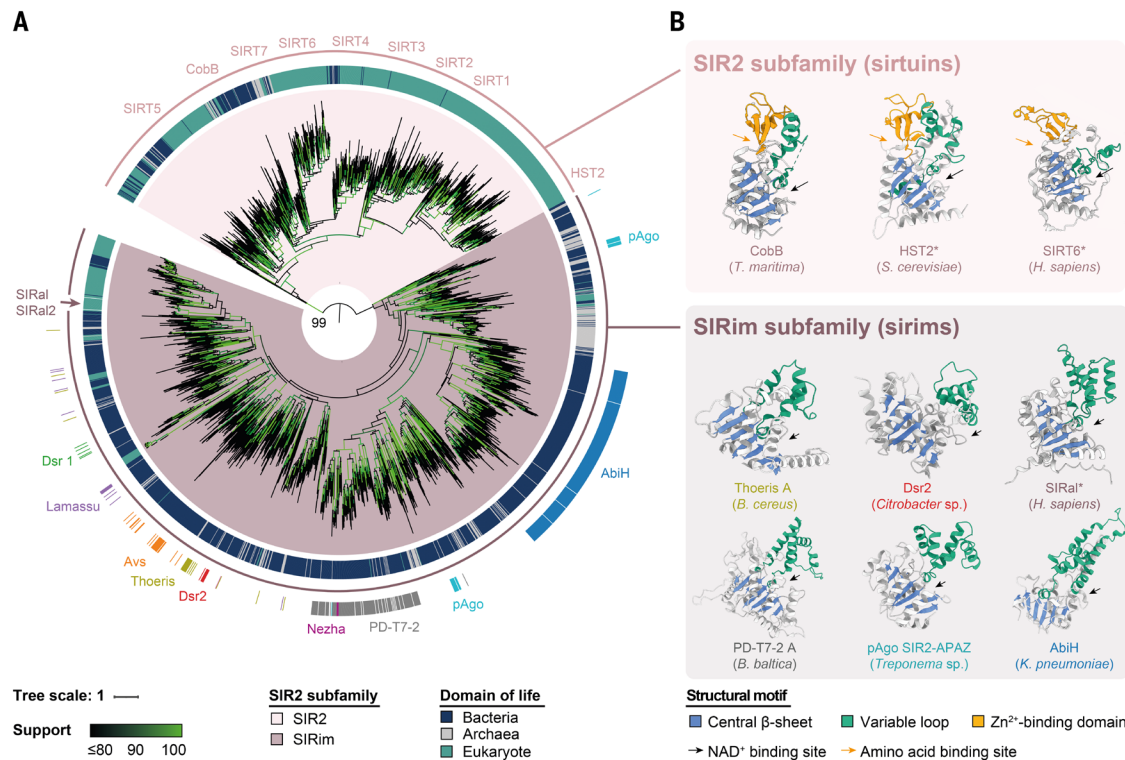


Fig. 1. Human proteins belonging to the monophyletic group of SIRim proteins are associated with bacterial antiphage defense proteins. (A) Phylogenetic tree of SIR2 family proteins across the tree of life. A total of 2723 sequences representative of all the SIR2 domain-containing proteins identified through a homology search were used to build the tree (midpoint rooted). SIR2 proteins cluster into two subfamilies based on amino acid sequences, namely SIR2 (inner ring, beige) and SIRim (inner ring, brown). The domain of life from which each protein originates is provided in the central ring (blue, bacteria; gray, archaea; green, eukaryotes). For proteins identified in bacterial organisms, we assessed whether the protein is present in a known antiphage defense system and indicated its name (outer ring) (21). The location of human sirtuins (SIRT1 to SIRT7), of SIRal and SIRal2, and of other proteins shown in (B) are indicated. Tree branch support values were assessed using UltraFast Bootstrap and are indicated in shades of green. Support is also specified on the branch delimiting the two subfamilies. **(B)** Structures of selected SIR2 proteins from the SIR2 subfamily (top, beige) and the SIRim subfamily (bottom, brown). Three sirtuins [CobB (PDB ID 2H2F), HST2 (PDB ID 1SZC), and SIRT6 (AFDB ID Q8N6T7)] and six sirims [Thoeris A (AFDB ID J8G6Z1, residues 1 to 279), Dsr2 (AFDB ID R8X3J8, residues 1 to 281), SIRal (AFDB ID Q9BPY3), PD-T7-2 A (AFDB ID I3Z638), pAgo SIR2-APAZ (EMBL ID AIW90156), and AbiH (AFDB ID A0A1V0M776)] are provided (PDB, Protein Data Bank; AFDB, AlphaFold Protein Structure Database; EMBL, European Molecular Biology Laboratory). For bacterial sirims, the name of the protein is colored according to the antiphage system in which the protein is found. In ThsA, Dsr2, and pAgo, only the SIRim domain is shown while it is fused to additional domains in the original proteins. The central β sheet of the Rossmann fold (blue) and the variable loop (green) are highlighted to underline the structural similarities between proteins. For each protein, the species of origin is indicated. An asterisk after a protein name indicates that it is found in a eukaryotic organism. Organisms featured are *Thermotoga maritima*, *Saccharomyces cerevisiae*, *Homo sapiens*, *Bacillus cereus*, *Citrobacter* sp. KTE151, *Belliella baltica*, *Treponema* sp. OMZ 838 and *Klebsiella pneumoniae*.

and predicted antiphage proteins, as well as in some eukaryotic proteins (Fig. 1A, brown).

Analyzing the experimental and predicted structures of SIR2 proteins sampled across the phylogeny revealed an overall conservation of the SIR2 domain topology, including the presence of a Rossmann fold with a three-layer sandwich. The central β sheet contains a core of six β strands, with accessory β strands in a subset of clades (Fig. 1B, blue). A variable α-helical loop is located downstream of the first β strand (Fig. 1B, green) (25). Sirtuins and sirims from bacteria, archaea, and eukaryotes show conservation of the NAD⁺ binding pocket, present at the junction between the Rossmann fold and the variable loop, as well as of catalytic residues involved in NAD⁺ degradation (fig. S2A) (16, 17, 20, 26). Structural differences, however, exist between sirtuins and sirims. The Rossmann fold of sirtuins, which bears a classical SIR2 domain, contains a β sheet with exactly six β strands, whereas seven or more strands are usually observed in sirim proteins. Sirtuins use a Zn²⁺ coordination site that allows binding of substrate amino acids of target proteins. Such a site is systematically absent from sirims, as is the tetrad of cysteines responsible for interactions with Zn²⁺ (Fig. 1B, orange, and fig. S2B). This suggests that proteins of the sirim clade are unable to bind acetylated amino acids, contrary to classical sirtuins

(16, 26). Overall, this structural and sequence analysis is congruent with the phylogenetic partition and suggests that the subfamilies of SIR2 proteins could have different functions.

We assessed the diversity of SIR2 proteins in humans by performing homology detection and identified nine proteins. Seven of these are sirtuins, well documented to regulate histone deacetylation (15, 22, 27). The other two, FAM118A and FAM118B, belong to the SIRim subfamily and have poorly documented functions, with no known immune role. Variations of FAM118A transcript levels have been observed in epithelium infected with respiratory syncytial virus, whereas FAM118B has been documented to play a role in Cajal bodies of the nucleus (28, 29). The evolutionary relationship between the FAM118 family of proteins and antiphage systems of bacteria led us to hypothesize that they may be involved in immunity. FAM118B was renamed SIRal (for SIR2 antiphage-like) and FAM118A SIRal2.

SIRal is involved in the mammalian TLR pathway

To gain insights into the function of SIRal, we analyzed the genomic localization of the *SIRal* gene. In bacteria, genes coding for proteins of a given pathway tend to colocalize within genomes, displaying functional synteny. Although less common, multiple occurrences of a functional

syntenic organization have been documented in eukaryotes (7–9, 30). In 73% of metazoans, including humans, *SIRal* is located two genes away from the gene that encodes TIRAP (fig. S3A). TIRAP carries a TIR domain and contributes to TLR2 and TLR4 signaling. Upon ligand-induced TLR activation, TIRAP's homotypic interaction with the cytosolic TIR domain of the receptor allows for immune signal transduction and production of proinflammatory cytokines (Fig. 2A) (31, 32). Synteny between the *SIRal* and *TIRAP* genes suggests that SIRal may be involved in the TLR pathway.

Analysis of publicly available datasets indicated that *SIRal* is broadly expressed in tissues and is enriched in macrophages (fig. S3B). The latter are essential immune cells that detect pathogens through pattern recognition receptors, including TLRs. TLR activation leads to inflammatory and antimicrobial responses, as well as to the priming of adaptive immunity. Cytokines such as tumor necrosis factor (TNF), interleukin-6 (IL-6) and IL-8, and interferons (IFNs) are produced upon TLR stimulation. IFN signaling through the IFN- α/β receptor (IFNAR) leads to the transcriptional up-regulation of antipathogen effectors called interferon-stimulated genes (ISGs). TLR signaling also results in the production of the antimicrobial agent nitric oxide (NO), which is synthesized by the enzyme inducible nitric oxide synthase (iNOS). To address the role of mouse SIRal (mSIRal) in the TLR pathway, we implemented a loss-of-function approach by knockout (deletion) in RAW264.7 murine macrophage-like cells (Δ mSIRal cells; fig. S3, C and D). The TLR response was assessed by stimulating cells with purified lipopolysaccharide (LPS) from *Escherichia coli*, an agonist of TLR4, and measuring the inflammatory response (Fig. 2, B and C). In wild-type macrophages, LPS stimulation led to up-regulation of iNOS and the production of NO, which was abrogated in Δ mSIRal macrophages (Fig. 2, B and C and fig. S3E). Δ mSIRal macrophages also displayed an impaired production of IL-6, IFN- β , and the ISG Rsad2 upon LPS stimulation, indicating that mSIRal is a central mediator of TLR4-driven inflammation (Fig. 2, B and C). By contrast, the production of TNF appeared mostly independent of mSIRal (Fig. 2, B and C). No defect in viability was observed for Δ mSIRal macrophages (fig. S3, F and G). Δ mSIRal cells expressed wild-type levels of TLRs, including TLR4, indicating that decreased NO production cannot be explained by receptor expression deficiency (fig. S3H). We additionally excluded LPS degradation in the absence of mSIRal by showing that both wild-type and Δ mSIRal cells trigger pyroptosis in response to cytosolic LPS sensing through inflammatory caspases (fig. S3G) (33). We further assessed the importance of SIRal by knockdown (reduced expression) in mouse and human primary macrophages (Fig. 2, D and E). Reducing mSIRal levels in mouse bone marrow-derived macrophages decreased IL-6 but not TNF production upon TLR4 stimulation (Fig. 2D and fig. S3, I and J). Human SIRal (hSIRal) knockdown in macrophages derived from blood translated into reduced expression of the ISGs Rsad2, IFI44L, and Mx1 upon LPS stimulation, whereas TNF and IL-8 production remained at control levels (Fig. 2E and fig. S3K). Altogether, these data demonstrate that SIRal plays a pivotal role in the induction of some inflammatory targets and ISGs downstream of TLR4 stimulation.

In addition to TLR4, it is well documented that other TLRs trigger the production of cytokines, IFN, and NO (34–36). To interrogate whether mSIRal is also involved in signaling downstream of these receptors, we stimulated wild-type and Δ mSIRal RAW264.7 cells with agonists of TLR1/2, TLR3, TLR7/8, and TLR9. In every case, a lack of mSIRal translated into impaired NO, IL-6, IFN- β , and Rsad2 production (Fig. 2F and fig. S3L). As observed with TLR4, a lack of mSIRal did not alter TNF production upon stimulation of the other members of the TLR family (Fig. 2F). Human primary macrophages knocked down for hSIRal similarly showed impaired cytokine and ISG up-regulation downstream of TLR7/8 stimulation (Fig. 2G and fig. S3, K and M). Transcription of inflammatory genes depends on activation of the transcription factor nuclear factor- κ B (NF- κ B) and interferon

regulatory factors (IRFs) by phosphorylation (Fig. 2A). LPS-driven NF- κ B and IRF3 phosphorylation was significantly affected in murine Δ mSIRal macrophages, suggesting that SIRal acts upstream of the nuclear translocation of these transcription factors (fig. S3, N and O). Innate immune activation depends not only on TLRs but also on other receptors such as cGAS that detect cytosolic DNA from viral or bacterial origin (37). cGAS produces the second messenger cGAMP that activates STING, leading to proinflammatory gene transcription. To assess whether hSIRal plays a part in other signaling pathways of innate immunity, we stimulated human primary macrophages with cGAMP and measured ISG expression (Fig. 2H). Knockdown of hSIRal resulted in a 20% decrease in Mx1 transcript, whereas levels of IL-8 and IFI44L transcripts remained unaffected (Fig. 2H). SIRal is thus not central to STING signaling. Similar results were obtained upon IFN signaling through the IFNAR receptor, indicating that SIRal's role in inflammation seems rather restricted to the TLR pathway (Fig. 2I). Altogether, these results document that SIRal is a central mediator of TLR-driven inflammation, demonstrating its role in mammalian immunity.

SIRal contributes to the control of HSV-1 and *Salmonella* in macrophages

TLRs protect against a variety of pathogens, including herpes simplex virus 1 (HSV-1), which is detected by TLR9, among other pattern recognition receptors, and the bacteria *Salmonella*, which is recognized by TLR4 (11). To test the importance of mSIRal in controlling pathogens that activate TLRs, we compared infection between control and Δ mSIRal RAW264.7 cells. By using a herpes virus strain engineered to express green fluorescent protein (GFP) fused to a viral capsid protein, it was possible to discriminate viral entry from productive viral replication. HSV-1 entry in cells resulted in intermediate GFP levels, whereas viral replication translated into the production of additional GFP molecules and cells with high GFP fluorescence (Fig. 3, A and B). Acyclovir, an inhibitor of viral replication, prevented the appearance of GFP⁺ high cells containing replicative virus (Fig. 3, A and B). Viral entry was equivalent in wild-type and Δ mSIRal cells (Fig. 3B and fig. S4, A and B). However, the absence of mSIRal translated into an increase in the percentage of productively infected cells and a twofold increase in GFP mean fluorescence intensity (MFI), a measure of viral replication (Fig. 3, B and C, and fig. S4, A to C). These results demonstrate that mSIRal plays a role in the control of HSV-1 infection in macrophages.

To interrogate the role of mSIRal in TLR-driven antibacterial immunity, RAW264.7 cells were infected with *Salmonella enterica* strain SL1344, and intracellular bacterial replication was measured by colony forming unit (CFU) after a gentamicin protection assay (Fig. 3D). A twofold increase in *S. enterica* accumulation was observed in Δ mSIRal cells compared with wild-type cells (Fig. 3D). Additionally, infected Δ mSIRal macrophages showed an abrogation of the production of the ISG Rsad2, whereas TNF transcript levels appeared unmodified (fig. S4D). To address whether TLR4-signaling is accountable for the difference of bacterial loads in wild-type and Δ mSIRal cells, we antagonized the receptor with LPS of *Rhodobacter sphaeroides* (Fig. 3E and fig. S4E) (38). Blocking TLR4 does not affect the accumulation of *S. enterica* in Δ mSIRal macrophages. Conversely, blocking TLR4 in wild-type cells translated into higher intracellular replication, reaching the levels measured in Δ mSIRal cells (Fig. 3E). Abrogating the difference between wild-type and Δ mSIRal cells by antagonizing TLR4 demonstrates that mSIRal acts within the TLR4 pathway in this infectious model (Fig. 3E). Altogether, these results show that mSIRal plays a role in TLR-driven immune protection against viruses and bacteria.

Sirims are widespread immune NADases among eukaryotes

Sirims from bacteria and animals are evolutionarily related proteins that participate in immunity. We investigated the evolutionary history of the SIRal lineage. To do so, we queried EukProt, a database containing protein sequences from 993 species spanning all major documented

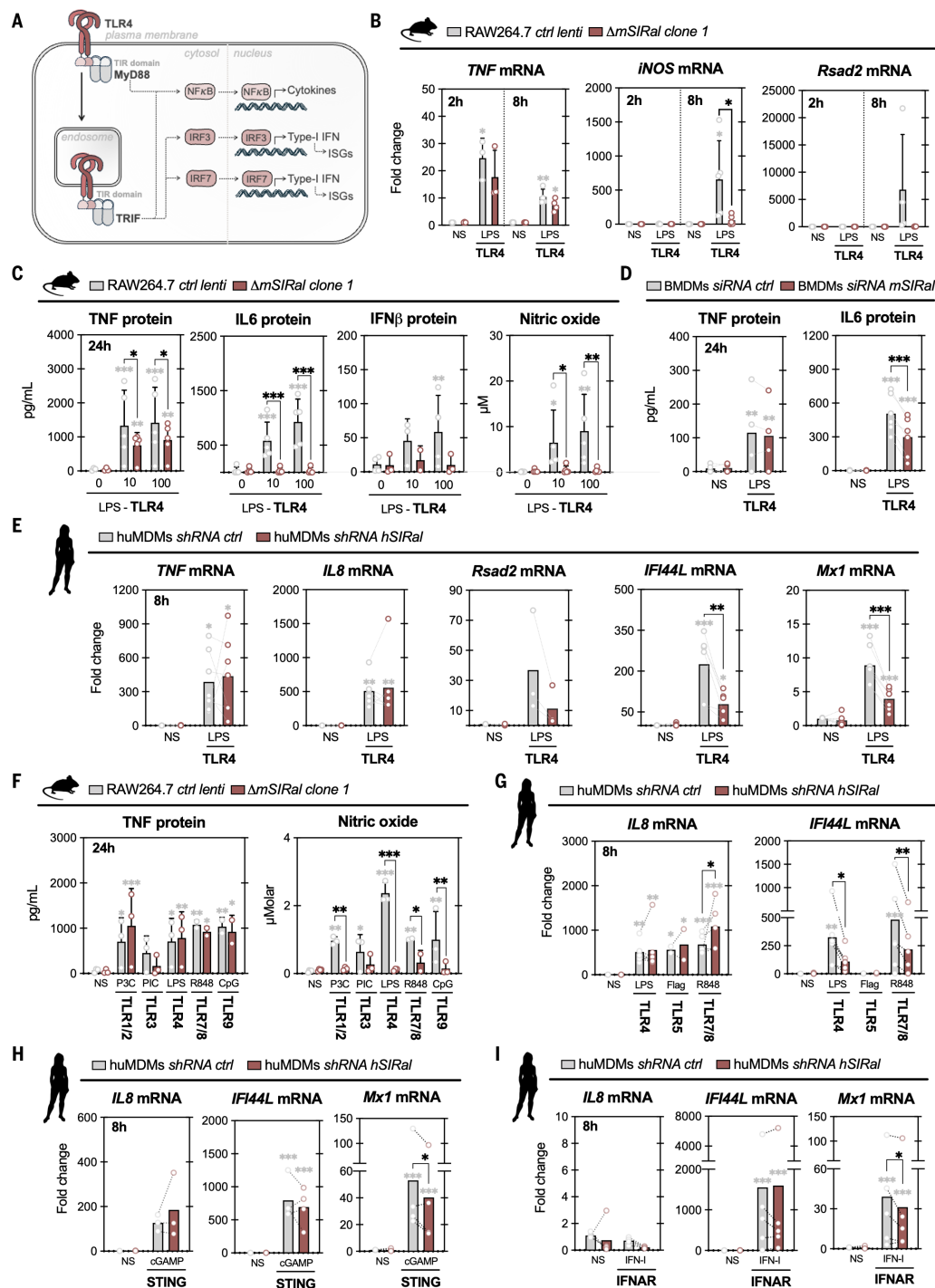


Fig. 2. SiRal contributes to TLR signaling in macrophages. (A) Schematic of TLR4 signaling in macrophages. (B) Wild-type or Δ mSiRal RAW264.7 cells were stimulated with LPS from *E. coli* (100 ng/ml). Levels of TNF, iNOS, and Rsad2 transcripts were measured by RT-qPCR (reverse transcription–quantitative polymerase chain reaction) at 2 and 8 hours. NS, not stimulated. (C) Wild-type or Δ mSiRal RAW264.7 cells were stimulated with LPS from *E. coli* (10 to 100 ng/ml). Production of TNF, IL-6, IFN- β , and NO was measured at 24 hours. (D) Mouse bone marrow-derived macrophages (BMDMs) transfected with a control small interfering RNA (siRNA) or a siRNA targeting mSiRal were stimulated with LPS from *E. coli* (100 ng/ml). Production of TNF and IL-6 was measured at 24 hours. (E) Human monocyte-derived macrophages (huMDMs) transfected with a control short hairpin RNA (shRNA) or a shRNA targeting hSiRal were stimulated with LPS from *E. coli* (100 ng/ml). Levels of TNF, IL-8, Rsad2, IFI44L, and Mx1 transcripts were measured by RT-qPCR at 8 hours. (F) Wild-type or Δ mSiRal RAW264.7 cells were stimulated with Pam3CysSK4 (500 ng/ml), Poly(I:C) (1 μ g/ml), LPS (100 ng/ml), R848 (1 μ g/ml), or CpG (1 μ g/ml), which are agonists of TLR1/2, TLR3, TLR4, TLR7/8, and TLR9, respectively. Production of TNF and NO was measured at 24 hours. (G) huMDMs transfected with a control shRNA or a shRNA targeting hSiRal were stimulated with LPS (100 ng/ml), flagellin (500 ng/ml), or R848 (1 μ g/ml), which are agonists of TLR4, TLR5, and TLR7/8, respectively. Levels of IL-8 and IFI44L transcripts were measured by RT-qPCR at 8 hours. (H and I) huMDMs transfected with a control shRNA or a shRNA targeting hSiRal were stimulated with either cGAMP (the agonist of STING, 4 μ g/ml) (H) or universal type I IFN (activator of IFNAR, 100 U/ml) (I). Levels of IL-8, IFI44L, and Mx1 transcripts were measured by RT-qPCR at 8 hours. Each dot represents the mean of technical replicates for each independent experiment [(B), (C), and (F)], independent human donor [(E), (G), (H), and (I)]. Bars correspond to the mean and error bars to the standard deviation of all the independent experiments, mice, or donors. Statistical analyses were performed using two-way analysis of variance (ANOVA); * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$, where gray asterisks correspond to comparisons within one group and black asterisks correspond to comparisons between groups.

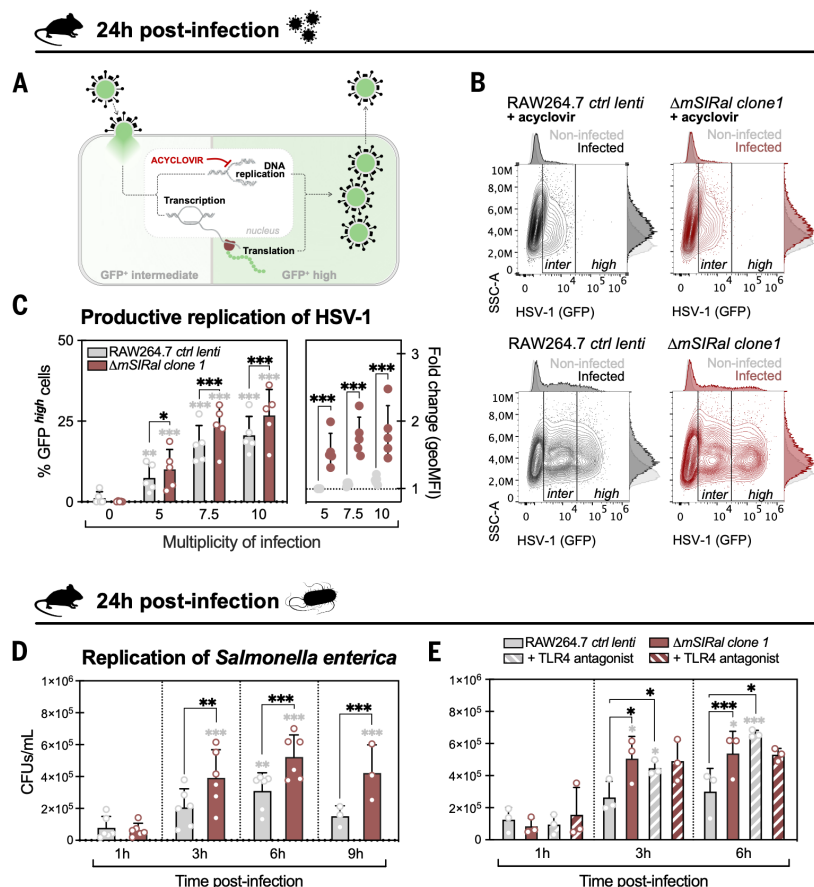


Fig. 3. mSIRal contributes to the control of HSV-1 and *Salmonella* in macrophages. (A) Schematic of HSV-1-GFP infection. Viral particles contain GFP fused to the capsid protein VP26, and hence, viral entry is detected in cells by intermediate GFP levels, as measured by flow cytometry. Productive viral replication results in GFP production and high levels of fluorescence. Viral replication, but not entry, is inhibited by the drug acyclovir. (B) Wild-type or ΔmSIRal RAW264.7 cells were either infected or not infected with HSV-1 [multiplicity of infection (MOI) 7.5] in the presence or absence of acyclovir. Levels of GFP were measured by flow cytometry at 24 hours. (C) Wild-type or ΔmSIRal RAW264.7 cells were either infected or not infected with HSV-1 (MOI 5 to 10). Levels of GFP were measured by flow cytometry at 24 hours, and the percentage of productively infected cells (GFP^{high}) was determined. The geometric mean fluorescence intensity (geoMFI) of the GFP^{high} population is indicated as a surrogate of viral replication. (D and E) Wild-type or ΔmSIRal RAW264.7 cells were infected with *S. enterica* serovar Typhimurium strain SL1344 (MOI 10). Intracellular replication of bacteria was quantified by CFU after gentamicin protection assay. In (E), LPS from *R. sphaeroides* (1 μg/ml) was used as a TLR4 antagonist. Each dot represents the mean of technical replicates for each independent experiment [(C), (D), and (E)]. Bars correspond to the mean and error bars to the standard deviation of all the independent experiments. Statistical analyses were performed using two-way ANOVA; **p* < 0.05, ***p* < 0.01, and ****p* < 0.001, where gray asterisks correspond to comparisons within one group and black asterisks correspond to comparisons between groups.

eukaryotic lineages, using the profile hidden Markov model (HMM) modeling SIRim proteins constructed in Fig. 1 (39). SIRal homologs were detected in a total of 35 diverse species, including a freshwater protist (*Potieriospumella lacustris*), zebrafish (*Danio rerio*), a slime mold (*Dictyostelium discoideum*), the Australian ghost shark (*Callorhynchus milii*), and the mallard (*Anas platyrhynchos*) (table S4). SIRal homologs are distributed in a wide range of eukaryotic taxa, suggesting that the acquisition in eukaryotic organisms is ancient (Fig. 4A and fig. S5, clade A). More precisely, the presence of SIRal homologs in many genomes of opisthokonta, a eukaryotic taxon that includes fungi, choanoflagellates, and animals (metazoa), suggests that the acquisition event occurred before their last common ancestor, more than 1.4 billion years

ago, and that major classes of opisthokonta such as insects have lost the gene (Fig. 4A, figs. S5 to S8, and table S4). Beyond the lineage of SIRal homologs (named clade A), our phylogenetic analysis uncovers four additional lineages of eukaryotic sirims (clades B to E) in 189 different species. In total, 19% of the eukaryotic species we queried encode a SIRim protein (Fig. 4A, figs. S5 to S7, and table S4). Clades A to C are distributed among a wide diversity of eukaryotes, with, for example, clade C encompassing 106 species. Conversely, clades D and E are limited to a small number of eukaryotes, with clade E detected only in choanoflagellates (Fig. 4A, fig. S5, and table S4). The interspersed distribution of the five eukaryotic lineages within bacterial and archaeal clades in the phylogenetic tree strongly suggests that the five clades of eukaryotic sirims arose independently during the history of eukaryotes. These lineages group together with bacterial sirims belonging to different antiphage systems (clades A to C with ThsA-like sirims, and clades D and E with pAgo-like sirims). These multiple acquisition events were followed by frequent gene losses, which explains the patchy distribution of sirims across eukaryotes (figs. S5 to S7).

We sought to gain insight regarding the biological function of eukaryotic sirims from the five identified clades. We observed that the SIRim domain is frequently fused to additional domains that are often observed in immune proteins. Clade A, for example, contains fungal and animal sirims with an N-terminal SIRim domain, a central nucleotide-binding domain (such as NACHT or NB-ARC), and C-terminal repeats (e.g., ankyrin or tetratricopeptide). This tripartite architecture is typical of NOD-like receptors (NLRs), which are immune receptors key to detecting a variety of infection cues in mammals and plants as well as to inducing programmed cell death upon mating of incompatible germlings in fungi (Fig. 4A, clade A) (40–43). In clade C, sirims are typically composed of a SIRim domain fused to a Macro domain (Fig. 4A). This is reminiscent of the domain architecture of ThsA from the type II Thoeris antiphage system in which a Macro domain is fused to an effector of cell death, a role that is fulfilled by a SIRim domain in type I Thoeris systems (44, 45). The genetic architectures of sirims thus suggest that they play an immune role in a wide diversity of eukaryotes.

To further probe this notion, we focused on the sirim homolog encoded in zebrafish (zSIRal, previously termed fam118b). Zebrafish encode for TLRs and display innate immune pathways that are evolutionary related to pathways present in mammals, including the production of ILs and IFNs as well as the transcriptional up-regulation of ISGs upon inflammation (46). To interrogate whether zSIRal is involved in innate immunity in vivo, we created crispants of the gene in zebrafish larvae (knockout approach) and performed infections with the *Edwardsiella tarda* strain FL60. No difference in survival was observed between wild-type and ΔzSIRal larvae upon infection (fig. S9A). We then assessed the inflammatory response upon infection by measuring IL and IFN transcripts 3 days postinfection. Infection triggered the up-regulation of ILs: *il26* in both genotypes and *il1β* in the ΔzSIRal larvae (Fig. 4C). However, the IFN response (*ifnγ* and *ifnφ1*) upon infection with *E. tarda* was visible in wild-type larva but abrogated in ΔzSIRal larvae (Fig. 4D), demonstrating that zSIRal is involved in regulating IFNs in vivo. Interestingly, ΔzSIRal larvae showed higher constitutive levels of *ifnφ1* than wild-type larvae (Fig. 4D).

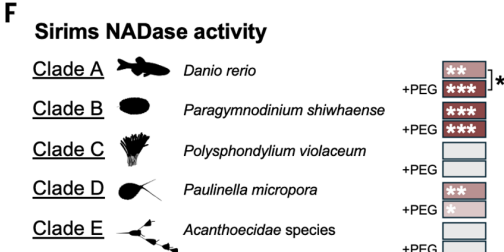
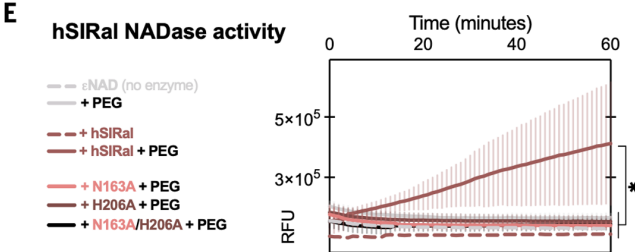
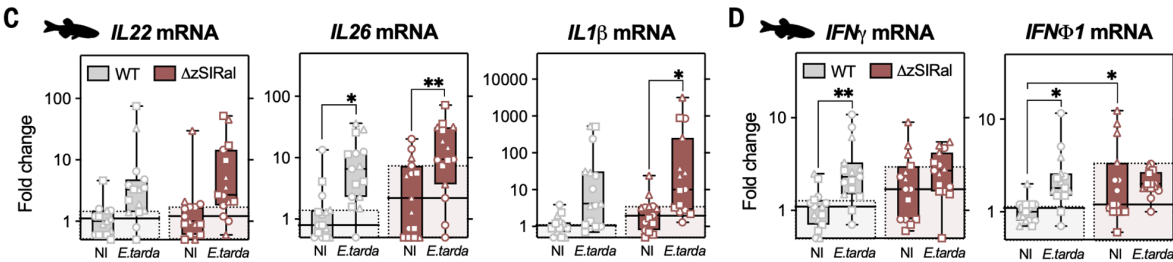
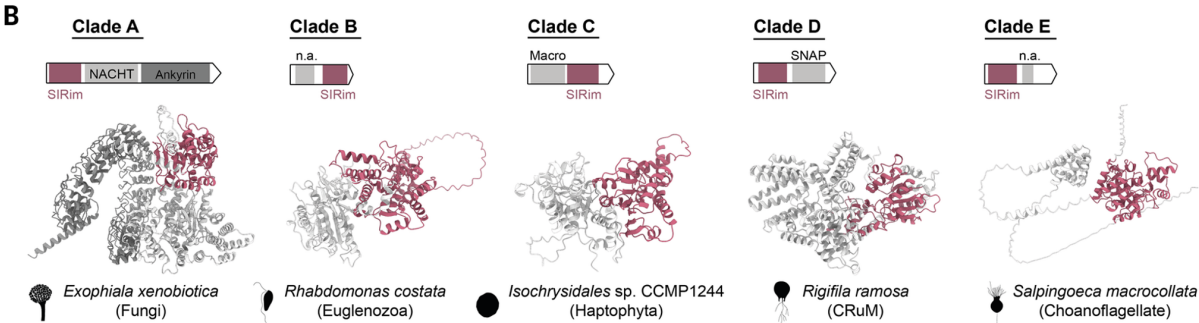
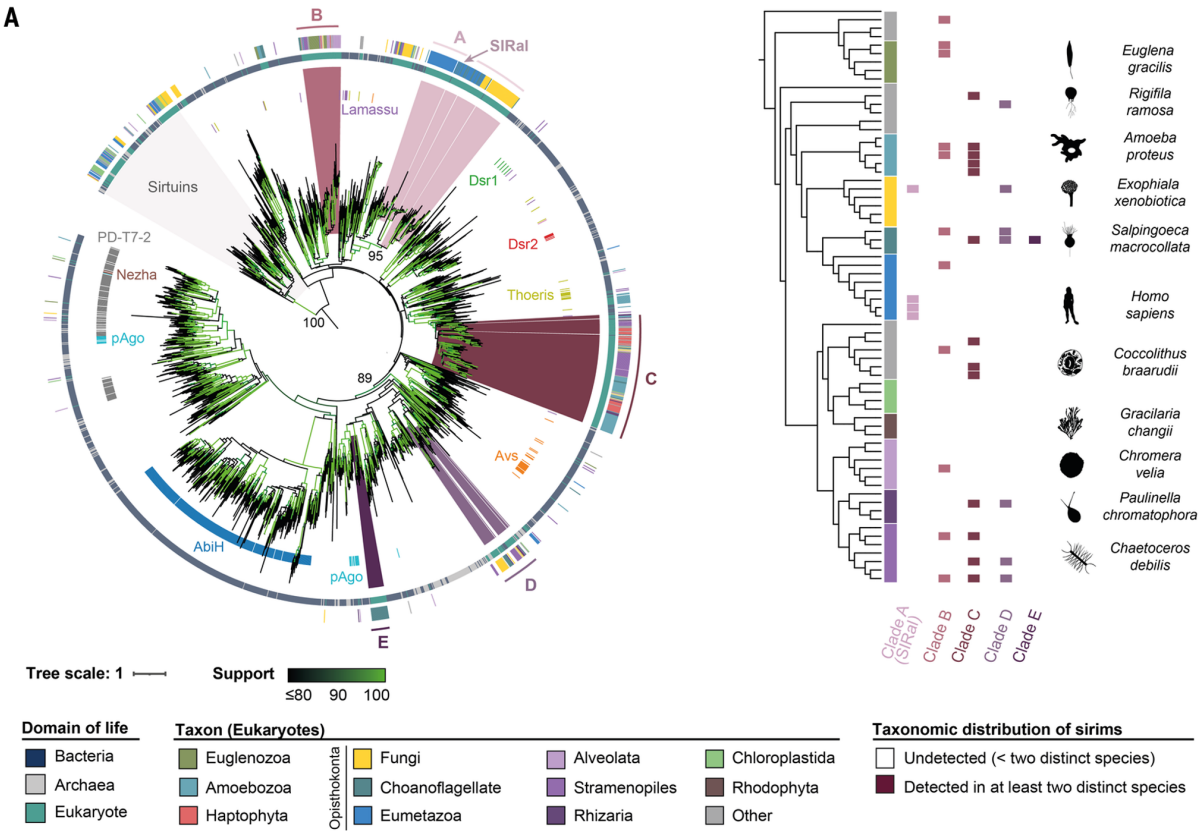


Fig. 4. SIRal is part of a large family of sirims in eukaryotes, including NADases with immune functions. (A) Shown on the left is a phylogenetic tree of all identified SIRim domain-containing proteins across the tree of life. Representative sequences of all detected bacterial, archaeal, and eukaryotic sirims ($n = 2266$) are aligned on their SIRim domain, and a phylogenetic tree was built from the multiple sequence alignment. The domain of life of each protein is indicated in the central ring (blue, bacteria; gray, archaea; green, eukaryotes), and the specific taxon is indicated for eukaryotic sequences in the outer ring. Antiphage systems were annotated using DefenseFinder for bacterial sirims (inner ring) (21). The five major eukaryotic clades of sirims (clades A to E) are colored in shades of purple (inner ring), and within each clade, specific sequences are detailed in (B). Branch support was assessed using UltraFast Bootstrap and is indicated in shades of green. Support values are indicated on key branches. Shown on the right is the taxonomic distribution of SIRim proteins in eukaryotes. The phylogenetic tree of The Comparative Set (TCS) from Eukprot includes 196 species from all major known eukaryotic phyla (39). The taxonomic distribution of each of the five eukaryotic SIRim protein clades is displayed. SIRim proteins detected in species present in the Eukprot v3 database (993 species) but not in TCS (196 species) were assigned to their closest relative present in the TCS tree. Sirim presence in a specific clade is reported only when at least two distinct species of the clade are identified as containing a *sirim* gene in their genome. The tree is midpoint rooted. The tree is also pruned, and branch lengths are ignored for visualization purposes (see fig. S5). For each major eukaryotic taxon, a logo of one example species encoding a *sirim* gene is provided. (B) Examples of sirims found in eukaryotic organisms. Sirims from each of the five major eukaryotic sirim clades are shown [(A) and figs. S5 to S7]. For each sequence, the domain architecture and the AlphaFold2 model are provided. The SIRim domain is colored in pink, and additional, fused, domains are indicated in gray. If the specified domain could be characterized using HHPred, its name is provided. NACHT (PF05729); SNAP (PF14938); n.a., domain not identified. (C and D) Wild-type or $\Delta zSIRal$ zebrafish larvae were infected with *E. tarda*. Levels of IL-22, IL-26, IL-1 β , IFN- γ , and IFN- ϕ 1 transcripts were measured by RT-qPCR at 3 days postinfection. Each symbol represents a larva, and bars correspond to the median of all the larvae. The three independent experiments are represented with dots, squares, and triangles. Statistical analyses were performed using two-way ANOVA; * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. NI, not infected. (E) Enzymatic NADase activity assay of purified recombinant hSIRal (wild type, N163A mutant, H206A mutant, N163A/H206A double mutant) incubated with ϵ -NAD with or without PEG 400 over time. RFU, relative fluorescence units. (F) Purified recombinant zSIRal from *D. rerio* (Opisthokonta, clade A), sirim from *P. shiwhaense* (Alveolata, clade B), *Polysphondylium violaceum* (Amoebozoa, clade C), *P. micropora* (Rhizaria, clade D), or *Acanthoeidae* sp. (Opisthokonta, clade E) were incubated with NAD $^{+}$ and with or without PEG 400. NADase activity was addressed by measuring NAD $^{+}$ degradation by HPLC analysis after 16 hours of incubation with NAD $^{+}$. Shades of red indicate statistical significance [p value notation is the same as in (D)].

Our results show that sirims function in both bacterial and eukaryotic immunity. We next investigated whether the underlying biochemical activity of sirims is conserved between bacteria and eukaryotes. Bacterial sirims perform an antiphage function by processing cellular NAD $^{+}$ (12, 20). To investigate whether SIRal similarly degrades NAD $^{+}$, recombinant hSIRal was purified and mixed in vitro with ϵ -NAD, an NAD $^{+}$ analog whose degradation can be monitored using a fluorescence-based assay. In the presence of ϵ -NAD alone, recombinant hSIRal did not demonstrate detectable ϵ -NAD degradation activity (Fig. 4E and fig. S9B). This finding, however, does not exclude the possibility that SIRal requires an activation trigger, such as a protein-protein interaction, to initiate NADase activity. In bacteria, activation of a homologous sirim, ThsA, depends on the oligomerization and formation of stable, enzymatically active protein filaments (20). We therefore hypothesized that hSIRal may be similarly activated through oligomerization. To assess this, we introduced the molecular crowding agent polyethylene glycol (PEG) as a means of artificially stimulating protein-protein interactions in vitro. In the presence of PEG, hSIRal demonstrated degradation of ϵ -NAD, albeit at a low rate (Fig. 4E). hSIRal enzymatic activity was further confirmed with the natural NAD $^{+}$ substrate and a high-performance liquid chromatography (HPLC) analysis, revealing reduced detection of NAD $^{+}$ and the presence of compounds whose elution and retention profiles are compatible with those of adenosine diphosphate ribose (ADPR) and nicotinamide (NAM) (fig. S9C). The NADase activity of bacterial ThsA can be abolished by introducing alanine substitutions for the conserved catalytic residues N112 and H152 (N, Asn; H, His) (12, 20). To generate a catalytically inactive hSIRal, we mapped these conserved residues on the human sequence, confirming the residues' putative homologous positions in an AlphaFold2 model, and constructed N163A (Asn to Ala at residue 163) and H206A (His to Ala at residue 206) single and double mutants (fig. S9, D and E). All predicted catalytically inactive hSIRal mutants had no detectable NADase activity, irrespective of the presence of PEG (Fig. 4E). To probe whether the NADase activity is conserved in diverse eukaryotic sirims, we subjected representative members of each clade to an HPLC-based NAD $^{+}$ in vitro cleavage assay. Sirims from clades A, B, and D showed interaction with NAD $^{+}$ (Fig. 4F and fig. S9, F and G). Sirims from mouse and zebrafish (clade A, mSIRal and zSIRal) as well as from the algae *Paragymnodinium shiwhaense* (clade B) cleaved NAD $^{+}$ into ADPR and NAM. The sirim from the amoeba *Paulinella micropora* (clade D) also led to a reduction in NAD $^{+}$ signal, without producing

detectable ADPR or NAM peaks. NADase activity was not detected for two sirims of clades C and E (Fig. 4F and fig. S9, F and G). In all cases, absence of the native activating signal or truncation of the protein to the SIRim domain alone may have also influenced NADase activity and still warrants further study. Overall, this demonstrates that the ability to hydrolyze NAD $^{+}$ is conserved in bacteria and eukaryotes and could be key for the immune function of sirims across domains of life.

We next evaluated whether the NADase activity is necessary for the immune function of eukaryotic sirims by investigating whether the role of hSIRal in the TLR pathway depends on its NADase activity. Human embryonic kidney cells (HEK 293T cells) engineered to express TLR4 and coreceptors, as well as a colorimetric reporter for NF- κ B-driven inflammation, can be used to monitor TLR4 signaling after LPS stimulation (fig. S9, H and I). Expression of wild-type hSIRal induced an average twofold increase in LPS-driven TLR4 inflammation, which was absent when the catalytically inactive version of hSIRal (N163A/H206A) was expressed instead (fig. S9, H and I). These results demonstrate that hSIRal participates in TLR signaling through its NADase activity. To interrogate whether the activation of SIRal is potent enough to deplete cellular NAD $^{+}$, we monitored intracellular NAD $^{+}$ concentration in murine macrophages upon LPS stimulation. $\Delta mSIRal$ macrophages did not show significant changes in NAD $^{+}$ concentration compared with control cells, suggesting that, as measured in vitro, SIRal is a weak NADase (fig. S9J). Our results demonstrate that akin to their bacterial homologs, eukaryotic sirims are NADases involved in immunity.

Discussion

In this work, we demonstrate that the widespread SIRim domain involved in antiphage defense also participates in eukaryotic immunity. We identified SIRal, a mammalian innate immune protein with a SIRim domain involved in signaling downstream of TLRs.

Given the vast body of literature on TLR pathways, one may be surprised that such a pivotal protein was not discovered sooner, through the implementation of a CRISPR screen or otherwise. One such CRISPR-Cas screen was performed to identify members of the TLR cascade but measured TNF levels as a readout of TLR4 activation (47). Our data indicate that its production is mostly independent of SIRal, which explains why it was not identified in that study. A recent publication also mapped members of the myddosome, a MyD88-dependent supramolecular structure responsible for signaling downstream of

TLR activation (48). SIRal was not identified in this work, suggesting that its mechanism of activation may be independent of interactions with myddosome components. SIRal was originally depicted as a protein involved in the Cajal bodies of the nucleus (28). Our data place it instead in the cytosol, where it participates in the TLR pathway. Its molecular mechanism is in the process of being deciphered.

The SIRim domain is involved in multiple antiphage systems, protecting bacteria against viral infection. In mammals, it is integrated within the TLR pathway and participates in protection against viral and bacterial infections. There is thus a diversification of this domain's protective role in eukaryotes. Conserved eukaryotic homologs of bacterial antiphage domains appear tailored to clade-specific immunity, congruent with the existence of selective pressures arising from eukaryote and/or clade-specific pathogens. Such adaptability was observed for additional conserved immune proteins, such as cGAS and its homologs coined cGAS-like receptors (cGLRs) and viperins (8, 9, 49). At the molecular level, the adaptation of the SIRim domain to the context of eukaryotic innate immunity translates into a shift in its biochemical activity: from a strong NADase in bacteria that depletes NAD^+ and triggers cell death to a weaker NADase activity in humans or zebrafish that does not trigger cell demise but probably participates in signal transduction.

Beyond humans and animals, we uncovered hundreds of sirims in a wide diversity of eukaryotes, including fungi, amoebas, alveolata, and diatoms. Other domains of antiphage origin also show conservation across a wide range of eukaryotes. TIR domains participate in immune signal transduction within the Toll receptor of insects and as part of mammalian TLRs (11). Homologs of cGAS and cGLRs are conserved across metazoans and participate in immunity in species such as coral and oyster (49–51). Based on SIRim-containing proteins being frequently fused with domains involved in immunity, and the ability of certain sirims to degrade NAD^+ , we predict that sirims could participate in host defense in many of these, often nonmodel, organisms. We have, for example, identified sirim homologs in *Chaetoceros* diatoms, which are keystone organisms in marine ecosystems. Studying immunity in these organisms could improve our understanding of how host-pathogen interactions shape ecosystems.

To account for the ever-growing number of immune members conserved across domains of life, we recently proposed the concept of ancestral immunity (52). Ancestral immune modules are proteins or protein domains that are evolutionarily conserved between prokaryotes and eukaryotes and involved in immunity in both domains of life. We show in this work that, akin to PNP or TIR, SIRim corresponds to an ancestral immune domain. Conservation across domains can be harnessed for mechanistic studies by interrogating the module's molecular features in multiple species in parallel (e.g., bacteria and humans). Whether SIRal plays a role in human pathology and can be harnessed for therapy, will be a topic of future investigation.

Materials and methods

Database of eukaryotic, archaeal, and bacterial genomes

We downloaded 4616 Eukaryotic genomes from 2407 species and 22,803 Bacterial complete genomes representing 7231 species from the GenBank database (<https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/>) in January 2022 (53). In addition, 2339 Archaeal genomes were also downloaded from the Genome Taxonomy DataBase (GTDB) in January 2023 and concatenated to them 496 inhouse Asgard Archaea genomes resulting in a database of 2835 Archaeal genomes (54). This resulted in a database of 30,254 genomes spanning across the three domains of life.

Systematic detection of SIR2 and SIR2-like proteins using iterative profile HMM search

An initial, loose, detection of SIR2 family proteins was performed by building a general profile hidden Markov model (HMM) modeling the SIR2 family of proteins. This was done by aligning known SIR2 homologs including (i) known eukaryotic homologs of sirtuins (e.g.,

orthologs of *Homo sapiens* SIRT1 to SIRT7) as well as (ii) homologs of bacterial antiphage proteins containing a SIR2 domain and identified by running DefenseFinder (v1.2.3, default parameters) on our database of genomes (21). SIR2 genes from the following systems were included in the selection of antiphage SIR2 proteins: *Thoeris*, *Dsr1*, *Dsr2*, *PD-T7-2*, *Nezha*, *Avs*, *AbiH*, *Lamassu* and prokaryotic *Argonautes* (pAgo). This yielded a set of $n = 4302$ sequences which were aligned using Muscle v5.1 (option -super5) and a profile HMM was built using the hmmbuild command from HMMER (version 3.3.2) (55, 56). This initial SIR2 family profile was queried against our database of genomes using hmmsearch (HMMER package) which yielded $n = 57,911$ hits satisfying the inclusion criteria (e-value $< 1e-5$, bit score > 20 , query coverage $> 60\%$ and $100 < \text{hit length} < 2500$ amino acids) in bacteria, archaea and eukaryotes (table S1). These proteins were clustered using MMseqs2 v1.3.45111 easy-cluster (30% identity, 80% coverage), representative sequences were aligned using Muscle v5.1 and the alignment was trimmed using Clipkit v1.3.0 (options -m gappy -g 0.99) (55, 57). An initial maximum likelihood phylogenetic tree was built from the trimmed alignment using IQTree v2.2.0 using the LG+G4 substitution model (58). Maximum likelihood optimization was performed for 1000 iterations (data not shown).

A refined detection of SIR2 proteins was then performed per SIR2 subfamily (SIR2 and SIRim separately) using an iterative profile HMM search against our database of bacterial, archaeal and eukaryotic genomes. This detection was performed to get a curated set of proteins per subfamily, to be used in all the remaining analyses. First, FAM118A and FAM118B orthologs were fetched from the NCBI Orthologs database (<https://www.ncbi.nlm.nih.gov/gene/79607/ortholog>), by filtering out proteins designated as “PREDICTED” or “LOW QUALITY” leading to a set of $n = 453$ FAM118 orthologs from Vertebrate organisms. Proteins were aligned using Muscle v5.1 (option -super5) and a profile HMM of FAM118A and FAM118B orthologs was constructed from the Multiple Sequence Alignment (MSA) using hmmbuild from the HMMER package (55, 56). This initial profile HMM was queried against the database of Eukaryotic proteins only in order to identify FAM118 homologs in this domain of life. This search (“Round 1”) led to obtaining $n = 751$ hits among Metazoa and Fungi satisfying the including criteria (e-value $< 1e-05$, $200 < \text{length} < 600$ amino acids). Hits were aligned using Muscle v5.1 (option -super5) and a new profile HMM was built to search the database of eukaryotic genomes for three additional rounds (“Round 2” to “Round 4”) using HMMER, until no additional SIR2-domain containing proteins were identified in eukaryotes (table S3) (55, 56). At each round, among all the profile HMM hits, only those satisfying the inclusion criteria were included in the list of valid hits and were included in the profile HMM for the subsequent profile HMM search (table S3). FAM118 homology search was then expanded in Bacteria and Archaea using four additional rounds of profile HMM search (“Round 5” to “Round 8”) against our database bacterial, archaeal and eukaryotic genomes. At each round, only hits satisfying the inclusion criteria were included to the list of valid hits (table S3). Iterative search was stopped once no additional hits satisfying the including criteria were retrieved from the database, yielding $n = 8697$ of sirims across domains of life. To this set of proteins were added the list of AbiH homologs ($n = 1208$) which were retrieved by querying the PF14253 Pfam profile against our database of genomes using hmmsearch (-cut_ga option) (56). This led to a total of 9905 hits including SIRal/SIRal2 (FAM118B/A) homologs in eukaryotes as well as proteins involved in known bacterial antiphage defense systems.

All deacetylating sirtuins were also retrieved from the database of genomes using a similar procedure of iterative profile HMM search. Initially, orthologs of all human sirtuins (SIRT1 to 7) were fetched on the NCBI Orthologs database and hits with “LOW QUALITY” or “PREDICTED” headers were filtered out. The set 3861 human sirtuin orthologs was clustered using MMseqs2 easy-cluster at 98% identity and 98% coverage, representative sequences were aligned using

Muscle v5.1 and a profile HMM of human sirtuin orthologs was built using hmmbuild (55, 56, 59). The obtained profile was iteratively queried against our database of eukaryotic genomes for three rounds ("Round 1" to "Round 3") until no novel eukaryotic sirtuins were detected (table S2). After round 3, hits satisfying the inclusion criteria (e-value < 1e-5, bit score > 110, query coverage > 80% and 200 < protein length < 1500 amino acids, $n = 23,192$) were clustered at 50% identity / 80% coverage using MMseqs2 easy-cluster. Representative sequences ($n = 1944$) were aligned using Muscle v5.1, a profile HMM was built using hmmbuild and the profile was queried against bacterial, archaeal and eukaryotic genomes. Iterative homology search was performed across the tree of life for two additional iterations until no additional sirtuin homologs were detected, leading to a set of $n = 46,623$ detected sirtuin homologs satisfying the inclusion criteria (e-value < 1e-5, bit score > 80, query coverage > 60% and 200 < protein length < 2500 amino acids (table S2).

Annotation of defense systems in bacterial genomes

To assess which SIR2 family proteins are involved in known antiphage systems, we extracted the genomic neighborhood (twenty genes upstream, twenty genes downstream) surrounding each sirtuin and sirim identified in the refined detection and ran DefenseFinder v1.2.3 with default parameters (Fig. 1A) (21). We also identified whether the genomic neighborhood of bacterial sirtuins and sirims is enriched in defense systems as antiphage defense systems tend to cluster together in bacterial genomes (23, 24). We measured whether at least one defense system (fig. S1A), as well as how many defense systems (fig. S1B), can be found in the genomic neighborhood of sirtuins and sirims (20 genes upstream and downstream). The difference in the frequency of neighboring defense systems between sirims and sirtuins was assessed using a two-sided Fisher exact test (fig. S1A) and the difference in the number of neighboring defense systems was assessed using a two-sided Mann-Whitney test (fig. S1B), as implemented in the scipy python package (v1.8.0) (60).

Phylogenetic inference

We first built a phylogenetic tree for the whole SIR2 family (including both SIR2 and SIRim subfamily proteins) using the proteins identified in the refined detection (Fig. 1A). All identified sirtuins ($n = 46,623$) and sirims ($n = 8697$) were clustered differentially in order to increase the representation of eukaryotic and archaeal proteins compared to their bacterial counterparts (59). Precisely, sirtuins were clustered at 50% identity-80% coverage, bacterial sirims were clustered at 50% identity-80% coverage whereas archaeal and eukaryotic sirims were clustered at 90% identity-90% coverage. A total of 2723 representative sequences ($n = 947$ sirtuins and $n = 1776$ sirims) were aligned on their SIR2 domain using Muscle v5.1 (option -super5 -perm none), the alignment was trimmed using Clipkit v1.3.0 (options -m gappy -g 0.99) and was used to build a maximum likelihood phylogenetic tree using IQTree (55, 59, 61, 62). The Q.pfam+G4 amino acid substitution model was selected using ModelFinder and the Bayesian Information Criterion (62). Maximum likelihood tree search was performed until convergence and branch support was assessed using 1000 UltraFast Bootstrap trees (63).

We then built phylogenetic trees of all sirims identified in our database of genomes (Fig. 4A and figs. S6 and S7). Prior to building a phylogenetic tree, in order to increase the representation of under sampled eukaryotic clades (e.g., rhizaria or alveolata), representative sequences of all identified sirims in the refined detection ($n = 8687$) were used to build a profile HMM that was queried against the Eukprot v3 database using hmmsearch (39). Hit satisfying the inclusion criteria (e-value < 1e-5, bit score > 55, query coverage > 60% and 100 < protein length < 2500 amino acids, $n = 482$ taking diverse isoforms of the same protein into account) among 205 genomes (over a total of 993 in the Eukprot v3 database) were further filtered to remove putative

bacterial contaminations. To do so, all $n = 482$ Eukprot v3 hits were used as queries with Diamond v1.2.6 (option -fast) against the NCBI nr database of proteins (64). Eukprot v3 proteins having more than 70% sequence identity with any bacterial protein in nr were filtered out. This led to a curated set of $n = 443$ sirims detected in $n = 189$ different genomes from the Eukprot v3 database (table S4). These curated hits were included in the set of all sirims used to build the tree. All sirims were clustered differentially depending on their domain of life using MMseqs2 easy-cluster: bacterial sirims were clustered at 50% identity-80% coverage whereas eukaryotic and archaeal sirims were clustered at 90% identity-90% coverage in order to increase their abundance in the tree compared to their bacterial counterparts (59). The set of representative sequences was manually cured by removing proteins lacking key amino acids in the GxG motif and in the xxNxD motif and gap-inducing proteins at these loci. Sirims found in the *H. sapiens* genome were explicitly added to the set of representative sequences. This step led to a total number of $n = 2101$ curated representative sequences of sirims. We also included a set of sirtuins from across the three domains of life to the tree of sirims to serve as an outgroup. To do so, $n = 165$ sequences were sampled from the set of representative deacetylating sirtuin sequences (clustered at 50% identity/80% coverage using MMseqs2 easy-cluster) were sampled from all three of domains of life and added to set of proteins included in the tree of sirims (59). The obtained representative sirtuins and sirims ($n = 2,266$) were then aligned on their SIR2/SIRim domain using Muscle v5.1 (option -super5), the MSA was trimmed using clipkit (option gappy, 99%) and a maximum likelihood phylogenetic tree was inferred using IQTree v3 (58). The Q.pfam+G4 substitution model was selected via ModelFinder using the Bayesian Information Criterion (BIC), and the tree was optimized until convergence and branch support was assessed using 1000 UltraFast Bootstraps (Fig. 4A) (62, 63). To assess if the topology of the phylogenetic tree of sirims is to perturbations of the input MSA, additional SIR2/SIRim domain MSA were generated using (i) MAFFT v7.505 L-INS-i algorithm (option -localpair) and (ii) Muscle v5.1 by permuting the guide tree when building the alignment (option -super5 -perm all). This procedure yielded five additional MSA of sirims which were trimmed using clipkit (-m gappy -g 0.99) and were used to build five phylogenetic trees with IQTree (same procedure as before) (55, 58, 61-63, 65) (fig. S6). Another phylogenetic tree containing only eukaryotic sirims was built (fig. S7). All detected eukaryotic sirims were clustered at 90% identity and 90% coverage using MMseqs2 leading to a set of $n = 485$ representative sequences which were aligned using Muscle v5.1 (option -super5) together with $n = 165$ sirtuins from across the tree of life serving as an outgroup to root the tree. The alignment was trimmed using Clipkit (-m gappy -g 0.995) and the tree was built using the Q.pfam+G4 substitution model and maximum likelihood search was performed until convergence (55, 58, 59, 62, 63). Tree branch support was assessed using 1000 generated UltraFast Bootstrap.

In all the phylogenetic trees of sirims, we looked for clades of eukaryotic sirims (Fig. 4A and figs. S5 to S7). The criteria to consider that a eukaryotic clade is valid are the following (i) the clade of eukaryotic proteins is well supported, (ii) the clade contains sequences from at least two distinct eukaryotic species, (iii) the genomes in which sirims are detected were produced in at least two distinct studies and (iv) the clades are stable to perturbations in the phylogenetic analysis (66). In the case where a clade of bacterial proteins forms a well-supported clade with the eukaryotic proteins and is basal to it, we consider that a putative Horizontal Gene Transfer (HGT) event from bacteria to eukaryotes is likely to have occurred (66).

To compute the taxonomic distribution of each lineage of sirims in eukaryotes, we analyzed the set of sirims detected in the Eukprot v3 database (genomes from $n = 993$ species). As no phylogenetic tree was available for the full Eukprot v3 species, we plotted the distribution on the phylogenetic tree of The Comparative Set (TCS) from Eukprot (genomes from $n = 196$ species) (39). For each species in Eukprot v3,

either the species is also present in TCS, or it is not. In the latter case, the species is assigned to its closest relative in Eukprot TCS using TaxonKit (67). For each species in TCS, we consider that sirims are detected if (i) sirims are detected in at least two Eukprot v3 different species represented by this species or (ii) the clade represented by TCS species contains only one Eukprot v3 species and sirims are detected within it (Fig. 4A, right, and fig. S5). Divergence time between eukaryotic taxa was taken from the TimeTree 5 (<http://www.timetree.org/>) (68).

Domain annotation and folding of eukaryotic sirims

We sought to annotate the protein domains of all detected eukaryotic sirims (clades A to E; Fig. 4B). Domain annotation was performed using the HHpred server (default parameters) against Pfam-A v36 and the Protein DataBank (PDB) mmCIF70 (69). Hits satisfying inclusion criteria (probability > 0.90) were considered valid. Proteins were also folded using AlphaFold2/ColabFold as implemented in the localcolabfold package v1.5.2 (<https://github.com/YoshitakaMo/localcolabfold>, single prediction using 12 optimization recycles) (70, 71).

RAW264.7 cell culture and agonist stimulation

RAW264.7 macrophage-like cells (initially purchased from ATTC) were kindly provided by C. Werts (Institut Pasteur, Paris, France). KO cell lines were generated using transduction with Lenti-CRISPR-v2 containing sgRNA targeting *mSIRal* / *fam118b* or non-targeting control guides (fig. S3C and table S5). Transduction was followed by puromycin selection, single cell cloning, and knock-out phenotype was confirmed using both PCR showing genomic deletions, and Western blot confirming the absence of protein expression (fig. S3, C and D). RAW264.7 cells (control and $\Delta mSIRal$ / *fam118b*) were cultivated in complete RPMI medium (Gibco) (10% FCS + Penicillin-Streptomycin) and were plated at 0.5×10^6 cells/ml in 200 μ l/96-well plate or 1ml/24-well plate (TPP) the day before stimulation. Stimulation with TLRs agonists was performed the day after plating. Cells were stimulated for 2 hours, 8 hours or 24 hours with LPS of *E. coli* strain O111:B4 (LPS-EB Ultrapure, Invivogen), synthetic triacylated lipopeptide (Pam3CysSK4, Invivogen), dsRNA analog poly(I:C) (PIC-HMW, Invivogen), flagellin of *Salmonella typhimurium* (FLA-ST Ultrapure, Invivogen), resiquimod (R848, Invivogen) or short synthetic single-stranded DNA (CpG ODN 1018, Invivogen). After stimulation, cells were frozen immediately for subsequent RNA extraction and supernatants were recovered for dosage of nitric oxide.

Primary mouse bone-marrow derived macrophages

Bone-marrow cells were obtained after cervical dislocation of female C57BL/6J mice (7-20 weeks old), extraction of tibias, femurs and iliac bones and bone marrow recovery by flushing with 22 G needle and syringe. Bone-marrow cells were differentiated during 7 days in complete RPMI medium (Gibco) (10% FCS + Penicillin-Streptomycin) supplemented with 10% L929 supernatant. After differentiation, bone-marrow derived macrophages (BMDMs) were scrapped in PBS and plated at 1×10^6 cells/ml in 200 μ l/96-well plate or 1ml/24-well plate (TPP) the day before transfection.

siRNA transfections of BMDMs

siRNA transfections were performed on differentiated BMDMs using lipofectamine reagent (Lipofectamine-3000, ThermoFisher) in serum-free medium (OptiMEM, Gibco). Briefly, pools of siRNA targeting *mSIRal* / *fam118b* (no. GS109229) (FlexiTube, Qiagen), or scrambled control siRNA (AllStar Negative Control, Qiagen) were incubated with lipofectamine in OptiMEM for 30 min at room temperature before drop-to-drop addition on the BMDMs at 100mM final concentration (4×25 nM of each siRNA). After 6 hours incubation with siRNA in OptiMEM, equivalent volume of complete medium was added and cell were incubated overnight for recovery and stimulated with agonists as described above.

Primary human monocyte-derived macrophages

Plasmapheresis residues from healthy adult donors were obtained from the Établissement Français du Sang (Paris, France) in accordance with the Institut National de la Santé et de la Recherche Médicale (France) (INSERM) ethical guidelines. According to French Public Health Law (art L 1121-1-1, art L 1121-1-2), written consent and Institutional Review Board approval are not required for human non interventional studies. Human peripheral blood mononuclear cells (PBMCs) were isolated from the blood of healthy donors using Ficoll-Paque (GE Healthcare) and CD14⁺ monocytes were then positively selected using magnetic beads (Miltenyi). Differentiation of the cells into macrophages was induced in RPMI medium (Gibco) with 5% FCS + 5% human serum AB (Sigma), penicillin-streptomycin (Gibco) with 50 ng/ml human macrophage colony-stimulating factor (M-CSF, Miltenyi) for 7 days. For cell transduction, 10×10^6 monocytes were plated in 10-cm non-treated cell culture dishes and treated with 5 ml of freshly collected lentiviral particles (see below), in the presence of 8 mg/ml of protamine sulfate (Sigma Aldrich). Selection of the transduced cells was performed with 2 μ g/ml of puromycin 48 hours post-transduction. Seven days post differentiation and transduction, human monocyte-derived macrophages (huMDMs) were collected, plated and stimulated with agonists as described above.

shRNA-lentiviral particles production for huMDMs

HEK293-LTV cells were maintained in DMEM supplemented with 10% FCS, 100 U/ml penicillin/streptomycin, and 1% non-essential amino acids (Gibco). Cells were routinely sub-cultured. For lentiviral production, 7×10^6 cells were seeded in T75 cm² flasks and transfected the next day. Transfection mixes consisted of 4.1 mg of lentiviral genome, 3 mg of the packaging plasmid PSPAX2 and 1.25 mg of VSV-G. DNA was complexed with 7.5 mM of polyethylenimine (PEI) for 20 min and added to cells. Sixteen hours later, culture media was renewed, and the lentiviral particles allowed to produce and accumulate for an additional 30 hours, after which they were collected, filtered on 0.45 μ m, and immediately used to transduce monocytes. The lentiviral vectors employed were pLKO.1-puro non-mammalian shRNA Control Plasmid DNA (SHC002, Sigma) and pLKO.1-Puro shRNA *hSIRal* / *fam118b* (TRCN0000130258, Sigma).

Cells RNA extraction and RT-qPCR analysis of cytokines and ISGs

RNA extraction was performed on frozen cells from p24-well plates using NucleoSpin RNA (Machery-Nagel), according to the supplier's instructions. Reverse transcription was performed with SuperScript RT (Invitrogen). RT-qPCR was performed with Taqman Universal Master Mix (Applied Biosystems), according to the manufacturer's instructions (references of primers in table S6). Fold change was calculated using the $2^{-\Delta\Delta Ct}$ method, with normalization by internal control (GAPDH for mouse cells and HPRT for human cells) followed by normalization by the non-stimulated condition.

NO dosage and cytokine quantification by ELISA

NO production by macrophages was measured in fresh macrophages supernatants from p96-well plates by the Griess reaction (Griess reagent system, Promega), according to the supplier's recommendations. Cytokine quantifications were performed on frozen cell culture supernatants from p96-well plates, using DuoSet ELISA kits (R&D) according to the manufacturer's instructions.

Propidium iodide integration and LDH release assay

RAW264.7 cells (control and $\Delta mSIRal$ / *fam118b*) were plated at 0.25×10^6 cells/ml in 200 μ l/96-well plate the day before transfection. Transfection with LPS of *E. coli* strain O111:B4 (LPS-EB, Invivogen) was performed using a transfection reagent (FuGENE, Promega) in OptiMEM (Gibco). Cells were then incubated overnight and LDH release in the supernatant was measured (CyQUANT cytotoxicity assay, ThermoFisher)

according to the manufacturer's instructions. In parallel, the integration of 1 µg/ml propidium iodide was measured by fluorescence on a plate reader (SpectraMax, MolecularDevices).

HSV-1 and *Salmonella* infections of RAW264.7 cells

RAW264.7 cells (control and Δ mSIRal / *fam118b*) were cultivated in complete RPMI medium (Gibco) (10% FCS + Penicillin-Streptomycin) and were plated at 0.5×10^6 cells/ml in 200µl/96-well plate or 1ml/24-well plate (TPP) the day before infection.

Viral infections were performed in 24-well plates, by addition of HSV-1 (GFP) in the cell culture medium at a multiplicity of infection (MOI) of 5, 7.5 or 10, with or without pre-incubation with 35 µM acyclovir. Cells were incubated for 24 hours, were detached using trypsin (TriPLE, Gibco) for 5 min at 37°C, were then stained for 5min with live/dead fixable viability dye in PBS (eFluor 450nm, eBioscience) before washing and 4% PFA-fixation during 10min. Cells were then resuspended in cytometry buffer (PBS, 3% FCS, 5mM EDTA, 0.02% sodium azide) and analyzed flow cytometry (CytoFLEX, Beckman Coulter). Analyses were performed on live cells (eFluor^{negative}) and both the percentage of GFP^{positive} (high and intermediate) cells and the geometric mean fluorescence intensity (gMFI) of the respective populations were analyzed (FlowJo v10, BD).

S. enterica serovar Typhimurium strain SL1344 used for bacterial infections was kindly provided by A. Rifflet (Institut Pasteur, Paris, France). Bacteria were grown overnight in liquid antibiotic-free LB medium for a maximum of 12 hours. Stationary bacterial culture was centrifuged at 4000 rpm for 10 min and resuspended in PBS before OD₆₀₀ measurement. Bacterial concentration was calculated based on standard curve with 1 OD₆₀₀ = 8×10^8 CFUs/ml. Infections of RAW264.7 cells were performed in 96-well plates by addition of bacteria at MOI 10 in antibiotic-free cell culture medium with or without 30 min pre-incubation with 1 µg/ml of LPS from *R. sphaeroides* (LPS-RS Ultrapure, Invivogen). After 30 min of infection, the cells were washed gently three times with PBS and incubated for 30 min in medium with 50 µg/ml gentamicin to kill all the extracellular bacteria. After gentamicin protection assay, cells were washed again three times with PBS and incubated in medium with 10 µg/ml gentamicin to prevent any extracellular replication of the residual bacteria. At the indicated time points (1-9 hours), media was removed and RAW264.7 cells were lysed with 0.2% sodium deoxycholate in PBS for 30 s. Intracellular bacteria were then enumerated by colony-forming unit (CFU) after plating on antibiotic-free LB-agar plates and overnight incubation at 37°C.

Protein extraction and Western blot analyses

RAW264.7 cells (control and Δ mSIRal / *fam118b*) were plated at 0.5×10^6 cells/ml in 3mL/6-well plate the day before stimulation. Cells were stimulated for 30 min or 3 hours with 100 ng/ml LPS of *E. coli* strain O111:B4 (LPS-EB Ultrapure, Invivogen). Cells were then scrapped, washed in PBS and lysed with RIPA buffer containing protease inhibitors (Roche) for 15 min on ice. After lysis, cells were centrifuged at full speed and the soluble fraction was recovered. Protein extracts were denatured with Laemmli + TCEP at 95°C for 10 min and loaded on 4%-15% gradient gels (BioRad) for SDS-PAGE analyses. After migration, the proteins were transferred on PVDF membrane (BioRad) using Trans-Blot Turbo (BioRad, mixed MW program). Membranes were then blocked in TBS with 0.1% Tween (TBS-T) and 5% non-fat milk. Probing of the membranes was performed overnight at 4°C with primary antibodies targeting phosphorylated proteins (p65 and IRF3). Membranes were then washed three times with TBS-T, incubated 1h at room temperature with HRP-coupled secondary antibodies and washed again three times with TBS-T. Revelation was performed using Immobilon Forte WB HRP-substrate (Millipore) using optimal auto-exposure on a ChemiDoc (BioRad). Finally, membranes were stripped 5 min using Restore PLUS buffer (ThermoFisher) and the same protocol was followed to monitor total proteins (p65 and IRF3). The following antibodies

were used: anti-phospho-p65 clone T.849.2 (ThermoFisher reference no. MA5-15160, 1:1000); anti-phospho-IRF3 clone 4D4G (CST reference no. 4947, 1:1000); anti-p65 clone 572 (ThermoFisher reference no. 436700, 1:1000); anti-IRF3 clone D83B9 (CST reference #4302, 1:1000), anti-mouse-IgG (H+L) (Jackson ImmunoResearch reference no. 115-035-003, 1:10 000); anti-rabbit-IgG (H+L) (Jackson ImmunoResearch reference no. 111-035-144, 1:10 000). All antibodies were used in TBS-T 1% non-fat milk. Quantifications of the Western blots were performed using ImageLab 6.1 software (BioRad) and the ratio (phosphorylated / total) was calculated for each protein after stain free normalization.

Plasmids

Plasmids were ordered from GenScript: pcDNA3.1 backbone (neomycin resistance) with human hSIRal / FAM118B, with a N-term FLAG-tag. Alanine substitution of hSIRal / FAM118B (N163A and H206A) was performed by site directed mutagenesis (Quick change XL, Agilent), according to the manufacturer's instructions. All Minipreps and Midipreps were performed using endotoxin-free kits (Macherey-Nagel).

TLR4–NF-κB reporter systems

HEK293T-Blue-Null2 and HEK293T-Blue-hTLR4 (initially purchased from Invivogen) were kindly provided by C. Werts (Institut Pasteur, Paris, France). NF-κB reporter HEK293T-Blue-Null2 and HEK293T-Blue-hTLR4 cells (stably transfected with human TLR4, MD2 and CD14) were cultivated in complete DMEM medium (Gibco) (10% FCS + Penicillin-Streptomycin). hSIRal / FAM118B overexpression was achieved by transfection of both cell lines with 10 ng/well pcDNA3.1-hSIRal / FAM118B (or catalytic mutant) in 12-well plates (TPP), using lipofectamine (Lipofectamine-3000, ThermoFisher) in serum-free medium (OptiMEM, Gibco). Transfection was performed for 48 hours, and cells were then collected by gentle flushing in PBS, enumerated and resuspended in commercial detection medium (HEK-Blue Detection, Invivogen), containing the substrate of the NF-κB reporter secreted embryonic alkaline phosphatase (SEAP). LPS stimulation was performed by incubating 50 000 transfected cells/well with 0.1-100 ng/ml LPS of *E. coli* strain O111:B4 (LPS-EB, Invivogen) overnight in detection media in 96-wells plate (TPP). Activity of the NF-κB reporter SEAP was measured by colorimetry at OD_{655 nm}.

Recombinant protein expression and purification

A synthetic gene construct (Integrated DNA Technologies) for hSIRal / FAM118B was cloned into a custom pET16 vector with an N-terminal 6xHis-hSUMO2 fusion tag, codon optimized for recombinant expression in *E. coli*, using Gibson assembly. N163A, H206A, and N163A/H206A double mutant hSIRal / FAM118B were all generated using 'round the horn' mutagenesis and the wild-type sequence as template. Sequencing results for plasmid preparations were confirmed before transformation into BL21-CodonPlus(DE3)-RIL *E. coli* (Agilent) for protein expression and subsequent purification as described below.

Transformant *E. coli* were plated on MDG-agar plates (0.5% glucose, 25 mM Na₂HPO₄, 25 mM KH₂PO₄, 50 mM NH₄Cl, 5 mM Na₂SO₄, 2 mM MgSO₄, 0.25% aspartic acid, 100 mg/ml ampicillin, 34 mg/ml chloramphenicol, and trace metals) and grown overnight at 37°C. Single colonies were then inoculated into liquid MDG media and grown overnight at 37°C, 230 rpm shaking. Overnight MDG cultures were used as inoculum for 1 liter M9ZB minimal media liquid cultures (0.5% glycerol, 1% Cas-amino Acids, 47.8 mM Na₂HPO₄, 22 mM KH₂PO₄, 18.7 mM NH₄Cl, 85.6 mM NaCl, 2 mM MgSO₄, 100 mg/ml ampicillin, 34 mg/ml chloramphenicol, and trace metals). M9ZB cultures were grown at 37°C, 230 rpm until OD_{600nm} >2.5, rested on ice for 20 min and then induced with 0.5 mM IPTG and set to shake at 230 rpm overnight at 16°C. Cells were pelleted by centrifugation, washed with PBS, and flash frozen in liquid N₂ for storage at –80°C.

Frozen cell pellets were resuspended in lysis buffer (20 mM HEPES-KOH pH 7.5 measured at room temperature, 400 mM NaCl, 10%

glycerol, 30 mM imidazole, 1 mM DTT) and subjected to sonication to extract soluble protein. Cell lysates were clarified with centrifugation and purified using conventional gravity-flow Ni-NTA affinity chromatography. Briefly, lysate was loaded on Ni-NTA resin (Qiagen) equilibrated with lysis buffer and the flow-through was added back over the column once to increase chances of binding. Columns were then washed with lysis buffer supplemented to 1M NaCl, washed with lysis buffer, and protein was ultimately eluted in lysis buffer supplemented to 300 mM imidazole. Eluted protein was dialyzed overnight (20 mM HEPES-KOH pH 7.5, 250 mM KCl, 5% glycerol, 1 mM DTT) in the presence of human SENP2 protease (D364–L589, M497A) to remove the N-terminal 6xHis-SUMO2 solubility tag. Protein was concentrated post-dialysis using 30 kDa cutoff Amicon centrifuge filters and loaded onto a Superose-6 16/600 (Cytiva) gel filtration column (equilibrated in 20 mM HEPES-KOH pH 7.5, 250 mM KCl, 1 mM TCEP) connected to an NGC FPLC (Bio-Rad). Fractions were verified for protein content and purity via SDS-PAGE analysis and pooled fractions were further concentrated >10 mg/ml and flash frozen with liquid N₂ for long-term storage at –80°C.

Fluorescence plate reader analysis of NADase enzymatic activity

Fifty-microliter reactions were assembled in PCR tubes before transfer to white-walled clear bottom 96-well plates. Reactions included hSIRal / FAM118B at 15 μ M, nicotinamide 1,N(6)-ethenoadenine dinucleotide (ϵ -NAD) at 500 μ M, 20 mM HEPES-KOH pH 7.5 at room temperature, 100 mM NaCl, and $\pm$ 10% PEG 400. All components were thawed on ice and reactions were incubated for 15 min at 37°C prior to addition of fluorescent NAD⁺ analog and transferred to a Synergy H1 plate reader (BioTek). Fluorescence recordings at 300 nm excitation/410 nm emission were measured in kinetic mode every minute for 1 hour.

HPLC analysis of SIRal and sirims enzymatic activities

Reactions included an individual sirim protein ranging from 3 μ M to 20 μ M, nicotinamide adenine dinucleotide (NAD⁺) at 500 μ M, 20 mM ammonium acetate pH 8.0 at room temperature, and $\pm$ 10% PEG 400. All components were thawed on ice. Then, 100 μ l reactions were assembled on ice and incubated 16 hours at 37°C. Samples were centrifuged at 13000 $\times$ g for 5 min at 20°C, and supernatants applied to 30 kDa cutoff Amicon centrifuge filters. Filtered flowthrough as well as adenine diphosphate ribose (ADPR) and nicotinamide (NAM) synthetic standards were then injected on the HPLC (Agilent 1200 Series) equipped with a 4.6 $\times$ 150 mm and 5 μ m particle-size Zorbax Bonus-RP C18 column, with absorbance at 254 nm collected over time. The HPLC program method used 0.1% (w/v) ammonium acetate running buffer with a methanol elution gradient of 0–100% over 10 ml, 100% methanol for 2 ml, followed by reequilibration in the running buffer. Elution times and peak integration corresponding to NAD⁺ were compared to standards to determine molecule identity and NAD⁺ depletion.

Zebrafish housing and husbandry

Zebrafish were maintained at a density of five adults per liter under controlled conditions, following European guidelines for zebrafish housing and care. The environmental parameters were as follows: photoperiod of 14 hours light / 10 hours dark, temperature of 26.5 $\pm$ 1°C, pH of 7.8 $\pm$ 0.1, conductivity of 240 $\pm$ 30 μ S/cm, and undetectable levels of NH₄⁺ and NO₂[–], with NO₃[–] concentrations below 50 mg/liter.

Embryos were incubated at 28°C until 5 days postfertilization (dpf). During the first 2 weeks of feeding, larvae were provided with rotifers (*Brachionus plicatilis*, >500 per fish per day), followed by dry food (Skretting, Gemma Micro) to apparent satiation, twice daily.

The AB zebrafish line used in this study was obtained from the European Zebrafish Resource Center (Germany). All procedures were conducted in accordance with European Union Directive 2011/63/EU and were approved by the local ethics committee (approval no. 59 CEEA).

Generation of SIRim crispants zebrafish

gRNA selection and preparation: Three guide RNAs (gRNAs) were designed to target the third exon of the SIRim gene for the generation of crispant larvae, following a protocol adapted from Shah *et al.* (72). Initially, seven candidate CRISPR RNAs (crRNAs) were selected using CRISPOR (<http://crispor.tefor.net/>) against the *D. rerio* UCSC May 2017 (GRCz11/danRer11) reference genome, using the 20bp-NGG – SpCas9, SpCas9-HF1, and eSpCas9 1.1 enzyme. Selection criteria included target locus proximity, specificity score, minimal off-target effects, and the absence of sequence motifs that could impair cleavage efficiency or specificity.

To generate gRNAs, 100 μ M of crRNA was annealed with 100 μ M of tracrRNA (Integrated DNA Technologies, IDT) by heating at 95°C for 5 min, followed by gradual cooling to 20°C. The gRNAs were then diluted in IDT duplex buffer to a final concentration of 20 μ M.

RNP complex formation and injection: Ribonucleoprotein (RNP) complexes were assembled using a mutated SpCas9 protein (R221K-N394K), generously provided by J.-P. Concordet (TACGene, MNHN). Each gRNA was combined with Cas9 at final concentrations of 13.3 μ M and 10 μ M, respectively. Individual gRNA-Cas9 complexes were injected separately into one-cell-stage zebrafish embryos to evaluate *in vivo* cleavage efficiency. To assess editing efficiency, pools of 10 larvae at 5 days dpf per gRNA injection were sequenced using Sanger sequencing (Eurofins Genomics, <https://eurofinsgenomics.eu/>). Sequence data were analyzed with the ICE online software (Synthego, <https://ice.editco.bio/#/>) to determine cleavage efficiency and predicted frameshift induction. Three gRNAs demonstrating >95% cutting efficiency and inducing 50–90% frameshift mutations through non-homologous end joining (NHEJ) and microhomology-mediated end joining (MMEJ) were selected for crispant generation.

Crispant generation: For the crispant experiments, each gRNA was assembled into RNP complexes separately to prevent preferential hybridization. The three RNP complexes were subsequently pooled at equimolar concentrations to yield a final concentration of 4.4 μ M per gRNA. A total of 2 nl of the final RNP mix was injected into one-cell-stage embryos. Control injections were performed using a mixture of the three negative control crRNAs (Alt-R Cas9 Negative Control crRNA nos. 1, 2, and 3; IDT), following the same assembly and injection protocol as the experimental groups.

crRNA sequences: The crRNA sequences targeting SIRim were obtained from IDT (SIRim crRNA 1: TCTGCAAGAGGACAAGAACC, SIRim crRNA 2: GACCAAGCGAGCTCCTGAGC, SIRim crRNA 3: CGCTGGCACTGAGGAGCTA). This experimental setup ensured robust gene disruption for subsequent phenotypic analyses of SIRim crispants.

E. tarda infection of wild-type and Δ SIRal zebrafish larva

E. tarda FL60, provided by P. Klesius (USDA, Agricultural Research Service, Aquatic Animal Health Research Unit), were grown in LB medium + 15 μ g/ml tetracycline at 28°C overnight. A 1:100 dilution was performed, and the bacteria were grown to reach OD₆₀₀ = 0.250. The bacteria were then centrifuged at 4000 rcf for 10 min and resuspended in E3 water to achieve OD₆₀₀ = 0.250. 40 ml of bacterial suspension in E3 were added to 40 larvae, 4 days dpf, for 5 hours at 28°C. Larvae were washed three times with E3 water. 3 days postinfection (dpi), infected larvae were used for qPCR.

Larva RNA extraction and RT-qPCR analysis of ILs and ISGs

Single larvae were used for RNA extraction. RNA was extracted using the Single Cell RNA Purification Kit (Norgen, no. 51800) following the manufacturer's instructions. Synthesis of cDNA was performed using the M-MLV Reverse Transcriptase Kit (Invitrogen, no. 28025013). qPCR was carried out using the Takyon Kit (Takyon, no. UF-LPMT-C0701) on

QuantStudio 5 System (sequences of primers in table S7). Fold change was calculated using the $2^{-\Delta\Delta Ct}$ method, with normalization by internal control (2f1a) followed by normalization by the wild-type non-infected condition.

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SUPPLEMENTARY MATERIALS

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Figs. S1 to S9; Tables S1 to S7; References; MDAR Reproducibility Checklist

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A human homolog of SIR2 antiphage proteins mediates immunity via the Toll-like receptor pathway

Delphine Bonhomme, Hugo Vaysset, Eirene Marie Q. Ednacot, Vasco Rodrigues, Yazan Salloum, Jean Cury, Anny Wang, Axel Benchetrit, Pierre Affaticati, Veronica Hernandez Trejo, Paul Vittot, Charlie Bories, Alexis Cornec, Jean-Pierre Levraud, Pedro P. Hernández, Philippe Benaroch, Benjamin R. Morehouse, Aude Bernheim, and Enzo Z. Poirier

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Editor's summary

Immune proteins can be conserved across domains of life, from bacteria to humans, although the extent of this conservation remains unclear. Bonhomme *et al.* identified a new protein family, the SIRims, by searching for eukaryotic proteins resembling bacterial immune proteins. They found that one member, SIRal, plays a key role in mammalian innate immunity by activating inflammation through the Toll-like receptor pathway. This work highlights how investigating conserved immune proteins across the tree of life can reveal previously unknown aspects of eukaryotic immunobiology, including in humans. —Di Jiang

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